

Morane-Saulnier Model N

French pilot Roland Garros was the first to fit a machine gun firing through the propeller to an aircraft in early 1915. He used steel deflector plates to prevent the bullets from damaging the blades. Following his success, the manufacturers produced about 50 Model Ns with similar fixtures. As it was one of the few armed fighters available, it was operated by several French and British units, but it was never popular, control being by old-fashioned wing warping rather than ailerons, and it was phased out by 1916.



ENGINE 80hp Gnome 9-cylinder rotary

WINGSPAN 8.1m (26ft 9in) **LENGTH** 5.8m (19ft 2in)

TOP SPEED 144kph (89mph) **CREW** 1

ARMAMENT 1 x .303in Lewis machine gun

Nieuport Type 17

Developed from the highly agile Type 11 "Bébé" (Baby) racer, that helped defeat the "Fokker Scourge" in 1916, the Type 17 offered superior performance and a synchronized machine gun on the upper fuselage.



ENGINE 110hp Le Rhone 9Ja 9-cylinder rotary

WINGSPAN 26ft 10in (8.2m) **LENGTH** 19ft (5.8m)

TOP SPEED 102mph (165kph) **CREW** 1

ARMAMENT 1 x .303in Vickers machine gun (firing through propeller); up to 8 Le Prieur rockets on interplane struts

Pfalz D.III

The Pfalz factory was in Bavaria, which was still a kingdom in its own right within the German Empire and wished to retain some control over its armed forces. Hence in the early war years, Pfalz built German and French planes under license for supply to Bavarian units only. In 1917, they brought out their own design, which was similar to the Albatros, with monocoque fuselage and Mercedes engine, but by no means a copy and in several respects superior.



ENGINE 160hp Mercedes D.III 6-cylinder water-cooled

WINGSPAN 30ft 10in (9.4m) **LENGTH** 22ft 10in (7m)

TOP SPEED 103mph (165kph) **CREW** 1

ARMAMENT 2 x 7.92mm Spandau machine guns

RAF S.E.5a



Designed around the Hispano-Suiza V8 engine, the S.E.5a was less maneuverable than the Camel, but faster, easier to fly, and a more stable gun platform. It could also sustain more battle damage and was the preferred mount of the great British aces, including Ball, Mannock, and Bishop. Its only problem was engine unreliability, solved by the introduction of the Viper V8.

ENGINE 200hp Wolsley W.4A Viper water-cooled V8

WINGSPAN 26ft 7in (8.1m) **LENGTH** 20ft 11in (6.4m)

TOP SPEED 120mph (193kph) **CREW** 1

ARMAMENT 1 x .303in Vickers machine gun (firing through propeller); 1 x .303in Lewis machine gun; 4 x 25lb (11kg) bombs

Sopwith Camel

The Camel was a development of the Pup, so-called because the tight grouping of the engine, guns, and pilot gave it a humped appearance. Formidable at height, and in particular in the right-hand turn, when the torque of the engine pulled it around amazingly swiftly, it needed a skillful and experienced pilot to fly it without getting into difficulty. Crashes in training were frequent. However, during the war, Camels shot down more enemy aircraft (over 1,200) than any other type.



ENGINE 130hp Clerget 9B 9-cylinder rotary

WINGSPAN 28ft (8.5m) **LENGTH** 18ft 9in (5.7m)

TOP SPEED 113mph (182kph) **CREW** 1

ARMAMENT 2 x .303in Vickers machine guns

Sopwith Pup

ENGINE 80hp Le Rhone 9C 9-cylinder rotary

WINGSPAN 26ft 6in (8.1m) **LENGTH** 19ft 4in (6m)

TOP SPEED 112mph (179kph) **CREW** 1

ARMAMENT 1 x .303in Vickers machine gun

Officially the Type 9901 in the RNAS and the Scout in the RFC, the "Pup" was universally known in both services by its nickname. This came about because it was a smaller version of the previous Sopwith model and because it was a delight to fly.



SPAD XIII



The great strength of this aircraft was in the powerful V8 engine designed by Marc Birgit of Hispano-Suiza, who was also responsible for the machine gun interrupter gear. During 1917, more were produced than any other Allied fighter type and it was extremely popular in the French fighting squadrons.

ENGINE 235hp Hispano-Suiza water-cooled V8

WINGSPAN 26ft 4in (8.1m) **LENGTH** 20ft 4in (6.2m)

TOP SPEED 139mph (222kph) **CREW** 1

ARMAMENT 2 x Vickers or Marlin machine guns

Vickers F.B.5 Gunbus

One of the first aircraft designed to fight; the F.B.5 used the layout of observation aircraft, to mount a forward firing machine gun. The less efficient pusher propellers combined with a low-powered engine and weight made for a slow and unresponsive airplane.



ENGINE 100hp Gnome Monosoupape 9-cylinder rotary

WINGSPAN 36ft 6in (11.1m) **LENGTH** 27ft 2in (8.3m)

TOP SPEED 70mph (113kph) **CREW** 2

ARMAMENT 1 x .303in Lewis machine gun